

Problem 1

A) No it is not a prisoners dilemma

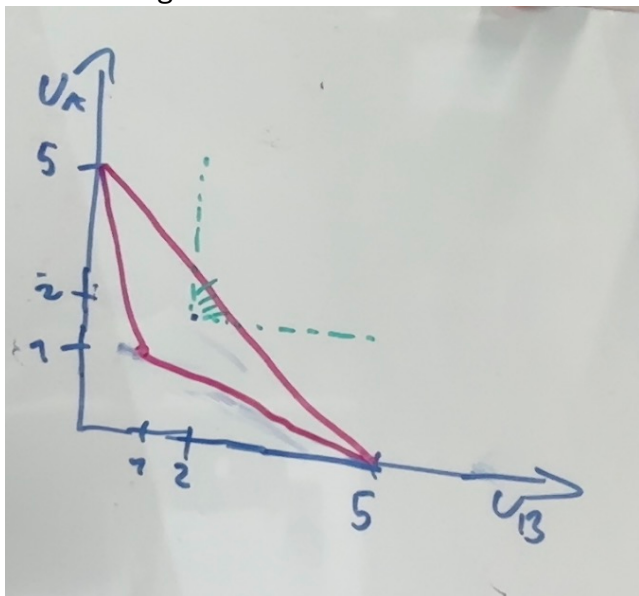
One might be tricked to thinking it is and it would be if the payoffs of $(leave, stay)$ and $(stay, leave)$ were switched

A prisoners dilemma would be if both players had a strictly dominating strategy and the payoff was not pareto optimal

In this case we see that both players have strictly dominating strategies in *leave*

Giving a unique NE: $(leave, leave)$ giving utility $(2,2)$ but there is no pareto improvement in the matrix

B) See drawing:



The convex hull of payoffs is drawn in pink while the average payoffs supported in SPNE for the infinitely repeated game is marked in green

Since $(2,2)$ is payoff in one repeated NE we must have $v > v^{NE}$ for v to be a SPNE

"supported" average payoff

Meaning that the "edge" in green is not part of the region where there are SPNE where this average payoff is reached

C) We use the One step deviation principal and since we are playing a symmetric(almost) game we just need to check for one player

Lets say player A

I calculate utility of deviation

$$u_A(\text{deviate}) = 2 + 5 \cdot \frac{\delta^2}{1 - \delta^2}$$

Then utility of non deviation

$$u_A(\text{dont deviate}) = 5 \cdot \frac{\delta}{1 - \delta^2}$$

When does no player want to deviate

Exactly when

$$u_A(\text{dont deviate}) \geq u_A(\text{deviate})$$

$$5 \cdot \frac{\delta}{1 - \delta^2} \geq 2 + 5 \cdot \frac{\delta^2}{1 - \delta^2}$$

$\Leftrightarrow$

$$\left(\delta \geq \frac{2}{3} \wedge \delta \leq 1 \right)$$

For $\delta \in \left[\frac{2}{3}, 1 \right)$ none of the players want to deviate giving us a SPNE in the specified strategy profile

D) It completely destroys the possibility of cooperation

In any finitely repeated game with a unique NE the unique NE is the only possible SPNE

Thinking about it:

In the last period a player will always play his best move since there are "no consequences" but since both players know this in the next to last round there are no consequences/punishment in the next to last round. This argument holds until the first game

In this specific situation the player B will always play his best move

Player A knows this meaning he will play his best move

Breaking the perfection of the earlier EQ

E) Player B still does not want to deviate as for $\delta \in \left[\frac{2}{3}, 1 \right)$

We simply need to check when player A wants to deviate

Again we calculate utility of deviation

$$u_A(\text{deviate}) = 2 + 4 \cdot \frac{\delta^2}{1 - \delta^2}$$

$$u_A(\text{dont deviate}) = 4 \cdot \frac{\delta}{1 - \delta^2}$$

When is it eq?

When he does not want to deviate

$$u_A(\text{deviate}) \leq u_A(\text{dont deviate})$$

$$2 + 4 \cdot \frac{\delta^2}{1 - \delta^2} \leq 4 \cdot \frac{\delta}{1 - \delta^2}$$

$\Leftrightarrow$

Uligheden løses for delta vha. WordMat.

$$\delta \leq -1 \vee \delta \geq 1$$

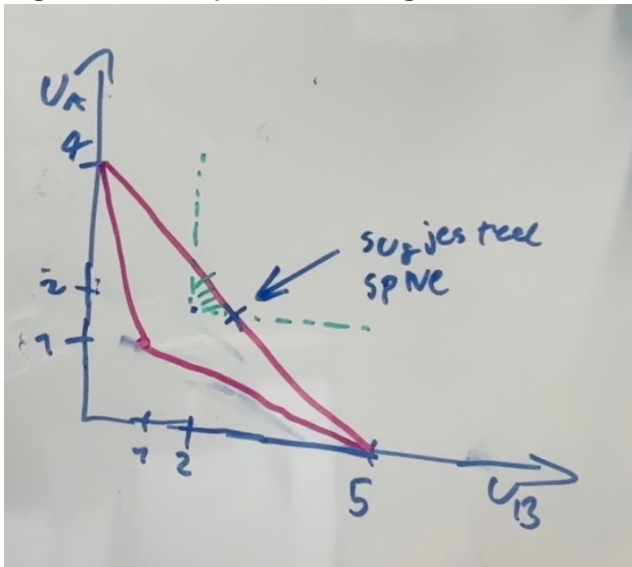
There are no $\delta \in (0,1)$ where he prefers to not deviate

Meaning that player A will always deviate meaning that the strategy in the game is never a SPNE

Looking back at the drawing from part b we also know that the average payoffs must have:

$v^* > v^{NE}$ for it to hold but this EQ gives player A 2 in average utility = 2

Which is not strictly better meaning that the average payoff vector is not in the “supported” region of the adjusted drawing



Problem 2

A) I check first separating then pooling

notation: (action by S^W , action by S^L) (action by $E|reg$, action by $E|adv$)

(reg, adv): E must play ($A|reg, R|adv$)

Will S deviate?

S^L prefers to deviate to reg getting $4 > 3$ utility

Meaning this is not an EQ

(adv, reg): E must play ($R|reg, A|adv$)

Will S deviate?

S^W gets highest possible utility meaning he will not deviate

S^L : by deviation the student gets utility $x = 2$ which is exactly what he is getting in EQ meaning he will not deviate

Thus we have a PBE in $\{(adv, reg)(R|reg, A|adv) \text{ for } \mu_{adv} = 1, \mu_{reg} = 0\}$

Pooling

(adv, adv) what does E play on path?

$$E[u_E(A|adv)] = \frac{1}{2} \cdot 4 = 2$$

$$E[u_E(R|adv)] = 1$$

$2 > 1$ thus E will play $A|adv$

Will S deviate?

S^W will never deviate since he has his highest payoff 5

S^L will deviate if E plays $A|reg$ so we need $E[u_E(A|reg)] \leq E[u_E(R|reg)]$

$$E[u_E(A|reg)] = \mu_{reg}$$

$$E[u_E(R|reg)] = (1 - \mu_{reg})$$

$$(1 - \mu_{reg}) \geq \mu_{reg}$$



Uligheden løses for μ_{reg} vha. WordMat.

$$\mu_{reg} \leq \frac{1}{2}$$

Thus we have a pooling EQ in

$$\{(adv, adv)(R|reg, A|adv) \text{ for } \mu_{reg} \leq \frac{1}{2}\}$$

(reg, reg): what will E play?

He doesn't care since

$$E[u_E(A|reg)] = \frac{1}{2} = E[u_E(R|reg)]$$

we have 2 possibilities ill investigate 1 by 1:

$A|reg$:

Will S deviate?

Yes S^W will always deviate since $5 > 3 > 2$

This actually also holds for $R|reg$ meaning that there are no pooling EQ in (reg, reg)

This means we have 2 PBE $\left\{ (adv, reg)(R|reg, A|adv) \text{ for } \mu_{adv} = 1, \mu_{reg} = 0 \right\}, \left\{ (adv, adv)(R|reg, A|adv) \text{ for } \mu_{reg} \leq \frac{1}{2} \right\}$

B) Does the pooling EQ survive the intuitive criterion?

Yes

S^W gets his highest possible payoff while S^L does not meaning we have to set $\mu_{reg} = 0$ but the EQ:

$$\left\{ (adv, adv)(R|reg, A|adv) \text{ for } \mu_{reg} \leq \frac{1}{2} \right\}$$

Is supported for $\mu_{reg} \leq \frac{1}{2}$ meaning $\mu_{reg} = 0$ supports it

Thus it survives the intuitive criterion

C) We saw earlier that the only possible pooling EQ is (adv, adv) since S^W always deviates from reg

E plays $A \forall x \in \mathbb{R}$ since it does not affect his payoff

S^W always gets his best payoff

Only one to possibly deviate is S^L

He will deviate $\forall \mu_{reg} \in [0, 1] \Leftrightarrow x < 2$

If E plays $A|reg$ he will deviate if $x < 4$

If E plays $R|reg$ he will deviate if $x < 2$

We can in this setting make beliefs such that an eq will hold

Meaning E will play R is reasonable

Thus we can find an eq if $x \geq 2$

And no pooling eq exists if $x < 2$

Problem 3

$$w_H = 100$$

$$w_L = 49$$

$$e \in \{0,1\}$$

$$p_1 = \frac{2}{3} = \text{probability of high income given high effort}$$

$$p_0 = \frac{1}{3} = \text{probability of high income given low effort}$$

$$d(0) = 0, d(1) = \frac{1}{3}$$

$$u(w, e) = \sqrt{w} - d(e)$$

A) I calculate the expected utility in each case

E=0:

$$\frac{1}{3} \cdot (\sqrt{w_H} - d(0)) + \frac{2}{3} \cdot (\sqrt{w_L} + d(0)) = \frac{1}{3} \cdot (10) + \frac{2}{3} \cdot 7 = 8$$

E=1:

$$\frac{2}{3} \cdot (\sqrt{w_H} - d(1)) + \frac{1}{3} \cdot (\sqrt{w_L} + d(1)) = \frac{2}{3} \cdot \left(\sqrt{100} - \frac{1}{3}\right) + \frac{1}{3} \cdot \left(\sqrt{49} - \frac{1}{3}\right) = \frac{26}{3} \approx 8.66666667$$

Meaning If no one is willing she will set $e = 1$

And get $\bar{u} = \frac{26}{3}$

B) Many parts: ill denote as I,ii,iii,iv,v

i) I write up the constraints in the second best (ill actually just write up the problem)

$$\begin{aligned} & \max_{B,p} p - \frac{2}{3} \cdot B \\ \text{s.t. } & \frac{1}{3} (\sqrt{w_H - p} - d(0)) + \frac{2}{3} (\sqrt{w_L + B - p} - d(0)) \geq \bar{u} \end{aligned} \quad (PC)$$

$$\begin{aligned} & \frac{1}{3} (\sqrt{w_H - p} - d(0)) + \frac{2}{3} (\sqrt{w_L + B - p} - d(0)) \\ & \geq \frac{2}{3} (\sqrt{w_H - p} - d(1)) + \frac{1}{3} (\sqrt{w_L + B - p} - d(1)) \\ & \Leftrightarrow \left(\frac{1}{3} - \frac{2}{3}\right) (\sqrt{w_H - p} - \sqrt{w_L + B - p}) \geq d(0) - d(1) \end{aligned} \quad (IC)$$

ii) In general in this setting we know that the first best solution will always work
So ill find the general one ignoring the IC for now and check it after

I find first best solutions

I write up the lagrangian where α_e is the probability of getting high wage given effort e

$$\mathcal{L} = p - (1 - \alpha_e)B + \lambda \left(\alpha_e (\sqrt{w_H - p} - d(e)) + (1 - \alpha_e) (\sqrt{w_L + B - p} - d(e)) - \bar{u} \right)$$

And find the FOC

$$\frac{\partial \mathcal{L}}{\partial p} = 1 - \lambda \left(\frac{\alpha_e}{2\sqrt{w_H - p}} + \frac{1 - \alpha_e}{2\sqrt{w_L + B - p}} \right) \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial B} = -(1 - \alpha_e) + \lambda \cdot \frac{1 - \alpha_e}{2\sqrt{w_L + B - p}} = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} \cdot \lambda = 0 \quad (3)$$

$$(2) \Leftrightarrow \lambda = 2\sqrt{w_L + B - p} > 0$$

I plug into 1

$$\begin{aligned} 1 - \lambda \left(\frac{\alpha_e}{2\sqrt{w_H - p}} + \frac{1 - \alpha_e}{2\sqrt{w_L + B - p}} \right) &= 1 - 2\sqrt{w_L + B - p} \cdot \left(\frac{\alpha_e}{2\sqrt{w_H - p}} + \frac{1 - \alpha_e}{2\sqrt{w_L + B - p}} \right) = 0 \\ \Leftrightarrow 1 &= \left(1 - \alpha_e + \alpha_e \cdot \frac{\sqrt{w_L + B - p}}{\sqrt{w_H - p}} \right) \\ \Leftrightarrow \frac{\sqrt{w_L + B - p}}{\sqrt{w_H - p}} &= 1 \Leftrightarrow \sqrt{w_L + B - p} = \sqrt{w_H - p} \Rightarrow B = w_H - w_L \end{aligned}$$

I plug into PC which is binding since $\lambda > 0$

$$\begin{aligned} \frac{1}{3}(\sqrt{w_H - p} - d(0)) + \frac{2}{3} \cdot (\sqrt{w_L + B - p} - d(0)) &= \bar{u} \\ \Leftrightarrow \sqrt{w_H - p} &= d(0) + \bar{u} \Leftrightarrow p = w_H - (d(0) + \bar{u})^2 \end{aligned}$$

We see if first best works for $e=0$ (if IC holds)

$$\left(\frac{1}{3} - \frac{2}{3} \right) (\sqrt{w_H - p} - \sqrt{w_L + B - p}) \geq d(0) - d(1)$$

Left side is 0 since

$$\sqrt{w_H - p} = \sqrt{w_L + B - p}$$

Left side side negative since $d(1) > d(0)$

Thus IC holds and FB is optimal with contract

$$B = w_H - w_L = 100 - 49 = 51$$

$$p = w_H - (d(0) + \bar{u})^2 = 100 - \left(\frac{26}{3} \right)^2 = \frac{224}{9} \approx 24,8888889$$

Giving the insurer expected profits

$$\Pi_{e=0}^{SB} = \Pi_{e=0}^{FB} = \frac{224}{9} - \frac{2}{3} \cdot 51 = -\left(\frac{82}{9} \right) \approx -9,1111111$$

This might indicate that ive made a mistake somewhere but ill just move on as if it should be negative as it makes snse on some level that its costly to implement the choice that isent natural for the Agent

iii) Again Ill just write up the problem

$$\max_{B,p} p - \frac{2}{3} \cdot B$$

$$s.t \quad \frac{2}{3}(\sqrt{w_h - p} - d(1)) + \frac{1}{3} \cdot (\sqrt{w_L + B - p} - d(1)) \geq \bar{u} \quad (PC)$$

$$\left(\frac{2}{3} - \frac{1}{3}\right)(\sqrt{w_h - p} - \sqrt{w_L + B - p}) \geq d(1) - d(0) \quad (IC)$$

Notice that its exactly the same except effort used and thus probabilities have changed

- iv) First we check if first best is viable then write up the lagrangian to show that both PC and IC must be binding and then find the optimal contract

Is first best viable: I.e does the IC constraint hold

$$\left(\frac{2}{3} - \frac{1}{3}\right)(\sqrt{w_h - p} - \sqrt{w_L + B - p}) \geq d(1) - d(0)$$

$$\sqrt{w_h - p} = \sqrt{w_L + B - p} \Rightarrow LHS = 0$$

$$d(1) - d(0) = \frac{1}{3} > 0 \Rightarrow IC \text{ broken}$$

I write up the lagrangian

$$\mathcal{L} = p - \frac{1}{3}B + \lambda \left(\frac{2}{3}(\sqrt{w_h - p} - d(1)) + \frac{1}{3} \cdot (\sqrt{w_L + B - p} - d(1)) - \bar{u} \right) \\ + \mu \left(\left(\frac{2}{3} - \frac{1}{3} \right) (\sqrt{w_h - p} - \sqrt{w_L + B - p}) - (d(1) - d(0)) \right)$$

I write up FOC

$$\frac{\partial \mathcal{L}}{\partial p} = 1 - \lambda \left(\frac{2}{3} \cdot \frac{1}{2\sqrt{w_h - p}} + \frac{1}{3} \cdot \frac{1}{2\sqrt{w_L + B - p}} \right) - \mu \left(\frac{2}{3} - \frac{1}{3} \right) \left(\frac{1}{2\sqrt{w_h - p}} + \frac{1}{2\sqrt{w_L + B - p}} \right) = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial B} = -\frac{1}{3} + \lambda \frac{1}{3} \cdot \frac{1}{2\sqrt{w_L + B - p}} - \mu \left(\frac{2}{3} - \frac{1}{3} \right) \cdot \frac{1}{2\sqrt{w_L + B - p}} = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} \cdot \lambda = 0 \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial \mu} \cdot \mu = 0 \quad (4)$$

We know that $\mu \geq 0, \lambda \geq 0$

AFC $\lambda = 0$

since $\mu \geq 0$ then (2) cannot hold since

$$\left(\frac{2}{3} - \frac{1}{3} \right) \cdot \frac{1}{2\sqrt{w_L + B - p}} > 0$$

Thus we must have $\lambda > 0$

AFC $\mu = 0$

But this gives us new FOC

$$\frac{\partial \mathcal{L}}{\partial p} = 1 - \lambda \left(\frac{2}{3} \cdot \frac{1}{2\sqrt{w_H - p}} + \frac{1}{3} \cdot \frac{1}{2\sqrt{w_L + B - p}} \right) = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial B} = -\frac{1}{3} + \lambda \frac{1}{3} \cdot \frac{1}{2\sqrt{w_L + B - p}} = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} \cdot \lambda = 0 \quad (3)$$

Which is exactly satisfied by of first best solution in which

$$\sqrt{w_h - p} = \sqrt{w_L + B - p}$$

Hut this weve shown break the the IC constraint meaning it is not viable

Thus this cannot be optimal and we must have

$$\mu > 0$$

Now that we know both are binding we have a “simple” system of equations

$$\frac{2}{3}(\sqrt{w_h - p} - d(1)) + \frac{1}{3}(\sqrt{w_L + B - p} - d(1)) = \bar{u} \quad (A)$$

$$\left(\frac{2}{3} - \frac{1}{3}\right)(\sqrt{w_h - p} - \sqrt{w_L + B - p}) = d(1) - d(0) \quad (B)$$

$$d(0) = 0, d(1) = \frac{1}{3}$$

$$(A) \Rightarrow \frac{2}{3}\sqrt{w_h - p} + \frac{1}{3}\sqrt{w_L + B - p} - \frac{1}{3} = \bar{u} \quad (C)$$

$$(B) \Rightarrow \frac{1}{3}(\sqrt{w_h - p} - \sqrt{w_L + B - p}) = \frac{1}{3} \Leftrightarrow \sqrt{w_h - p} - \sqrt{w_L + B - p} = 1$$

$$\Leftrightarrow \sqrt{w_h - p} = \sqrt{w_L + B - p} + 1 \quad (D)$$

Plug into C

$$\frac{2}{3} \cdot (\sqrt{w_L + B - p} + 1) + \frac{1}{3}\sqrt{w_L + B - p} - \frac{1}{3} = \bar{u}$$

$$\sqrt{w_L + B - p} + \frac{1}{3} = \bar{u} \Leftrightarrow w_L + B - p = \left(\bar{u} - \frac{1}{3}\right)^2 \quad (E)$$

$$\Leftrightarrow B = \left(\bar{u} - \frac{1}{3}\right)^2 - w_L + p \quad (F)$$

$$(E) + (D) \Rightarrow \sqrt{w_h - p} = \bar{u} - \frac{1}{3} + 1 = \bar{u} + \frac{2}{3} \Leftrightarrow w_H - p = \left(\bar{u} + \frac{2}{3}\right)^2 \Leftrightarrow p = w_H - \left(\bar{u} + \frac{2}{3}\right)^2$$

Giving us compensation scheme

$$B = \left(\bar{u} - \frac{1}{3}\right)^2 - w_L + p$$

$$p = w_H - \left(\bar{u} + \frac{2}{3}\right)^2$$

I plug in numbers first finding p then B

$$p = 100 - \left(\frac{26}{3} + \frac{2}{3}\right)^2 = \frac{116}{9} \approx 12,8888889$$

$$B = \left(\frac{26}{3} - \frac{1}{3}\right)^2 - 49 + \frac{116}{9} = \frac{100}{3} \approx 33,3333333$$

Giving expected profits

$$\Pi_{e=1}^{SB} = \frac{116}{9} - \frac{1}{3} \cdot \frac{100}{3} = \frac{16}{9} \approx 1,7777778$$

v) Remembering

$$\Pi_{e=0}^{SB} = \Pi_{e=0}^{FB} = \frac{224}{9} - \frac{2}{3} \cdot 51 = -\left(\frac{82}{9}\right) \approx -9,1111111$$

We see that

$$\Pi_{e=0}^{SB} = -9,1111111 < 1,7777778 = \Pi_{e=1}^{SB}$$

Thus the insurance company prefers to simply cover it all by implementing the the SB solution with high effort

And compensation scheme

$$p = w_H - \left(\bar{u} + \frac{2}{3}\right)^2 = \frac{116}{9} \approx 12,8888889$$

$$B = \left(\bar{u} - \frac{1}{3}\right)^2 - w_L + p = \frac{100}{3} \approx 33,3333333$$

C) Pareto domination is when one or more agents are better off without anyone being worse off anyone has at least as much utility and some have higher

In this EQ we see that the doctor has the exact same expected utility, since PC is binding and the insurance company has strictly higher utility since the company gets 0 if the doctor is not insured

Problem 4

Im assuming V_i are common knowledge

A) The demand function looks something like

$$Q(p) = \begin{cases} 1 & \text{for } p \leq v_L \\ \lambda & \text{for } p \in (v_L, v_H] \\ 0 & \text{for } p > v_H \end{cases}$$

The cinema will maximize profits choosing either

$$p = v_L$$

Giving

$$\pi_L = v_L = 4$$

Or

$$p = v_H \\ \pi_H = \lambda \cdot v_H = \lambda \cdot 12$$

C will only sell to enthusiasts $\Leftrightarrow \pi_H > \pi_L \Leftrightarrow 12 \cdot \lambda > 4 \Leftrightarrow \lambda > \frac{1}{3}$

B) Again I use I,ii,iii

i) I write it up

$$\max_{p_1, p_2} p_1 \cdot \lambda + \delta \cdot (1 - \lambda) \cdot p_2$$

s. t

$$v_H - p_1 \geq 0$$

(PCH)

$$v_L - p_2 \geq 0$$

(PCL)

$$v_H - p_1 \geq \delta(v_H - p_2)$$

(ICH)

$$\delta(v_L - p_2) \geq v_L - p_1$$

(ICL)

ii) I argue

ICL + ICH

$$v_H - p_1 \geq \delta(v_H - p_2) \Leftrightarrow v_H(1 - \delta) \geq p_1 - \delta p_2$$

$$\delta(v_L - p_2) \geq v_L - p_1 \Leftrightarrow p_1 - \delta p_2 \geq v_L(1 - \delta)$$

Together

$$v_H(1 - \delta) \geq p_1 - \delta p_2 \geq v_L(1 - \delta)$$

This implies $p_1 \geq p_2$

From this

PCL+ICH implies PCH

$$0 \leq v_L - p_2 < v_H - p_2 \\ \Rightarrow \delta(v_H - p_2) \geq 0 (> \text{for } \delta \neq 0) \\ v_H - p_1 \geq \delta(v_H - p_2)$$

Giving us remaning constranints

$$v_L - p_2 \geq 0$$

(PCL)

$$v_H - p_1 \geq \delta(v_H - p_2)$$

(ICH)

$$\delta(v_L - p_2) \geq v_L - p_1$$

(ICL)

ICH must bind

AFC ICH slack

$$v_H - p_1 > \delta(v_H - p_2)$$

Increasing p_1 means higher profits and doesn't violate any other constraints

Meaning it cannot be optimal

thus

$$v_H - p_1 = \delta(v_H - p_2)$$

ICL is slack

From before we have

$$v_H(1 - \delta) \geq p_1 - \delta p_2 \geq v_L(1 - \delta)$$

Binding ICH gives

$$\begin{aligned} v_H - p_1 = \delta(v_H - p_2) &\Leftrightarrow p_1 - \delta p_2 = v_H(1 - \delta) \geq v_L(1 - \delta) \quad (> \text{for } \delta \neq 0) \\ &\Leftrightarrow p_1 - \delta p_2 \geq v_L(1 - \delta) \Leftrightarrow \delta(v_L - p_2) \geq v_L - p_1 \end{aligned}$$

Thus ICL is implied by the others

Remaining constraints

$$v_H - p_1 = \delta(v_H - p_2) \quad (ICH)$$

$$v_L - p_2 \geq 0 \quad (PCL)$$

PCL must bind otherwise increase p_2 by ϵ and p_1 by $\delta \cdot \epsilon$ increasing profits

Thus PCL must bind

This gives us a system of 2 equations to solve

$$v_L - p_2 = 0 \Leftrightarrow p_2 = v_L$$

$$\begin{aligned} v_H - p_1 &= \delta(v_H - p_2) \Leftrightarrow v_H - p_1 = \delta \cdot (v_H - v_L) \\ &\Leftrightarrow p_1 = v_H - \delta(v_H - v_L) = v_H(1 - \delta) + v_L \end{aligned}$$

Plugging in numbers

$$\begin{aligned} p_2 &= v_L = 4 \\ p_1 &= 12(1 - \delta) + 4 = 16 - 12\delta \end{aligned}$$

iii)

the "separating option" gives profits

$$\Pi_a = \lambda(16 - 12\delta) + \delta(1 - \lambda)4 = 16\lambda - 12\lambda \cdot \delta + 4\delta - 4\lambda\delta = 16\lambda(1 - \delta) + 4\delta$$

Selling in opposite way first to non enthusiasts then enthusiasts doesn't make sense as they'll simply take the first ticket always

Selling in one day must always be on the first since the second is strictly worse

Thus we must sell everything in the first period if not separating

Thus we have the 2 other options

Selling to everyone on day 1

$$\Pi_L = 4$$

Selling to enthusiasts on day 1

$$\Pi_H = \lambda \cdot 12$$

Now what is best depends on delta

$$\Pi_H = \lambda \cdot 12$$

When is solution from “a” best?

$$16 \cdot \lambda \cdot (1 - \delta) + 4 \cdot \delta \geq 4$$

$\Updownarrow$

$$\left(\frac{1}{4} \leq \lambda \quad \wedge \quad 1 - \delta \geq 0 \right)$$

For $\lambda \geq \frac{1}{4}, \delta \leq 1$

$$16 \cdot \lambda \cdot (1 - \delta) + 4 \cdot \delta \geq \lambda \cdot 12$$

$\Updownarrow$

$$(1 - 4 \cdot \delta) \cdot \lambda + \delta \geq 0$$

$$\delta \geq \lambda \cdot (1 - 4\delta)$$

$\Updownarrow$

$$\lambda \geq \frac{1}{4}, (4 \cdot \delta - 1) \cdot \lambda + \delta \geq 0$$

When is solution π_H best

Form earlier

$$\lambda \geq \frac{1}{3}$$

$$16 \cdot \lambda \cdot (1 - \delta) + 4 \cdot \delta \leq \lambda \cdot 12$$

$\Updownarrow$

Uligheden løses for lambda vha. WordMat.

$$(4 \cdot \delta - 1) \cdot \lambda \geq \delta$$

$$\pi_H \text{ optimal} \Leftrightarrow \lambda \geq \frac{1}{3}, (4 \cdot \delta - 1) \cdot \lambda \geq \delta$$

π_L optimal:

Anywhere else ☺

$$16 \cdot \lambda \cdot (1 - \delta) + 4 \cdot \delta \leq 4$$

$\Updownarrow$

Uligheden løses for lambda vha. WordMat.

$$\left(\lambda \leq \frac{1}{4} \quad \wedge \quad 1 \geq \delta \right)$$

From before

$$\lambda \leq \frac{1}{3}$$

Thus we just need

π_L optimal for $\lambda \leq \frac{1}{4}$

- C) Well it depends on some factors especially how patient people are especially if we started to differentiate between different patience levels for each type it might become a lot better but since players can simply wait it becomes maybe a little weak since the waiting also hurts the firms profits
as it stands now I don't think its especially good